

MARCO STEFANELLI

Meteorological and Oceanographic Data Scientist | Numerical Weather Prediction | Oceanographic modelling | Data Assimilation | Artificial Intelligence

- Nationality: Italian
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- Personal webpage: <https://stfmrc.github.io/MarcoStefanelli/>
- Languages: Italian (native), English (fluent), Croatian (beginner)

PROFESSIONAL SUMMARY

Climate, Meteorological and oceanographic data scientist with a PhD in Earth System and Climate Science and hands-on experience in climate modelling, numerical weather prediction (NWP), oceanographic modelling, model evaluation, data assimilation, improvements in severe precipitation forecasting, artificial intelligence, and Python scientific computing. Strong background in climate, atmospheric and ocean modelling, large-scale meteorological data analysis, Linux/HPC workflows, and communicating results clearly to interdisciplinary teams. Currently developing AI-enhanced NWP and data assimilation methods to improve severe precipitation forecasting using weather radar observations.

CORE COMPETENCIES

Climate modelling | Oceanographic modelling | Numerical Weather Prediction (NWP) | Forecast validation and model evaluation | Severe precipitation and weather radar-based forecasting | Data assimilation | AI applications | Python (xarray, NumPy, PyTorch, TensorFlow, geopandas, rasterio) | Linux/Unix | HPC (SLURM, PBS, LSF) | Fortran | NetCDF/GRIB/HDF5 | Satellite and in situ observations.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher, SMASH (Marie Skłodowska-Curie COFUND), Faculty of Mathematics and Physics, University of Ljubljana | Apr 2024 – Apr 2026

- Developing AI-based data assimilation methods to improve weather forecasting, with a focus on severe precipitation events.
- Designing a neural-network observation operator for weather radar data assimilation.
- Working with large meteorological datasets, NWP model outputs, and forecast workflows in Python and Linux/HPC environments.
- Translating research outcomes into operationally relevant improvements.
- Communicate research outcomes clearly to interdisciplinary teams.

Researcher, Euro-Mediterranean Centre on Climate Change (CMCC), Italy | Feb 2023 – Mar 2024

- Advanced ocean-model validation products against observations
- Developing ocean data assimilation methodologies for unstructured grid modelling.

Researcher, The Abdus Salam International Centre for Theoretical Physics (ICTP), Italy | Apr 2018 – Apr 2019

- Investigate the effects of urbanisation on the climate of the West African Monsoon.
- Applied atmospheric and climate modelling approaches to analyse land-atmosphere interactions.

EDUCATION

- **PhD, Future Earth, Climate Change and Societal Challenges — University of Bologna & CMCC, Italy** / Nov 2019 – Jan 2023
Dissertation: Data assimilation for advanced cross-scale unstructured-grid ocean modelling.
Abstract: <https://amsdottorato.unibo.it/id/eprint/10785/>
Download: https://amsdottorato.unibo.it/id/eprint/10785/1/Thesis_Stefanelli.pdf
Related publication: <https://doi.org/10.3389/fmars.2025.1656879>
- **MSc, Earth System Physics — University of Trieste, Italy** / Sep 2014 – Mar 2018
Dissertation: A study of urbanisation effects on the West-African monsoon.
- **BSc, Physics — University of Salento, Italy** / Oct 2009 – Dec 2013
Dissertation: Mean sea level processes, observations, and evolution in the Mediterranean Sea.

SELECTED PUBLICATIONS

- Stefanelli, M., Jansen, E., Aydogdu, A., Federico, I., Coppini, G. (2025). Data Assimilation for Advanced Cross-Scale Unstructured-Grid Ocean Modelling. *Frontiers in Marine Science*.
<https://doi.org/10.3389/fmars.2025.1656879>
- Stefanelli, M., Zaplotnik, Z., Skok, G. Preprint: A Neural Network-Based Observation Operator for Weather Radar Data Assimilation (under review).
<https://doi.org/10.48550/arXiv.2512.18289>

SELECTED PRESENTATIONS

- European Geosciences Union (EGU), Wien (2022 and 2025)
2022 - <https://meetingorganizer.copernicus.org/EGU22/EGU22-4741.html>
2025 - <https://meetingorganizer.copernicus.org/EGU25/EGU25-420.html>
- International Conference on Alpine Meteorology (ICAM) (Poreč, Croatia) (2025)
- European Meteorological Society (EMS) Annual Meeting, Ljubljana (2025)
Video: <https://meetingorganizer.copernicus.org/EMS2025/EMS2025-28.html>
- Slovenian Meteorology Society, Ljubljana (2025)
Video: <https://www.youtube.com/watch?v=kIYwkgOSsGA>

ADDITIONAL INFORMATION

Grants: Marie Skłodowska-Curie COFUND postdoctoral grant (SMASH) | Puglia Region Youth Fund (PIN).

Tools and data: NetCDF libraries (NCO, CDO, NCAP), Jupyter, GitHub, GRIB/HDF5/NetCDF, weather radar, satellite, and in situ observations.